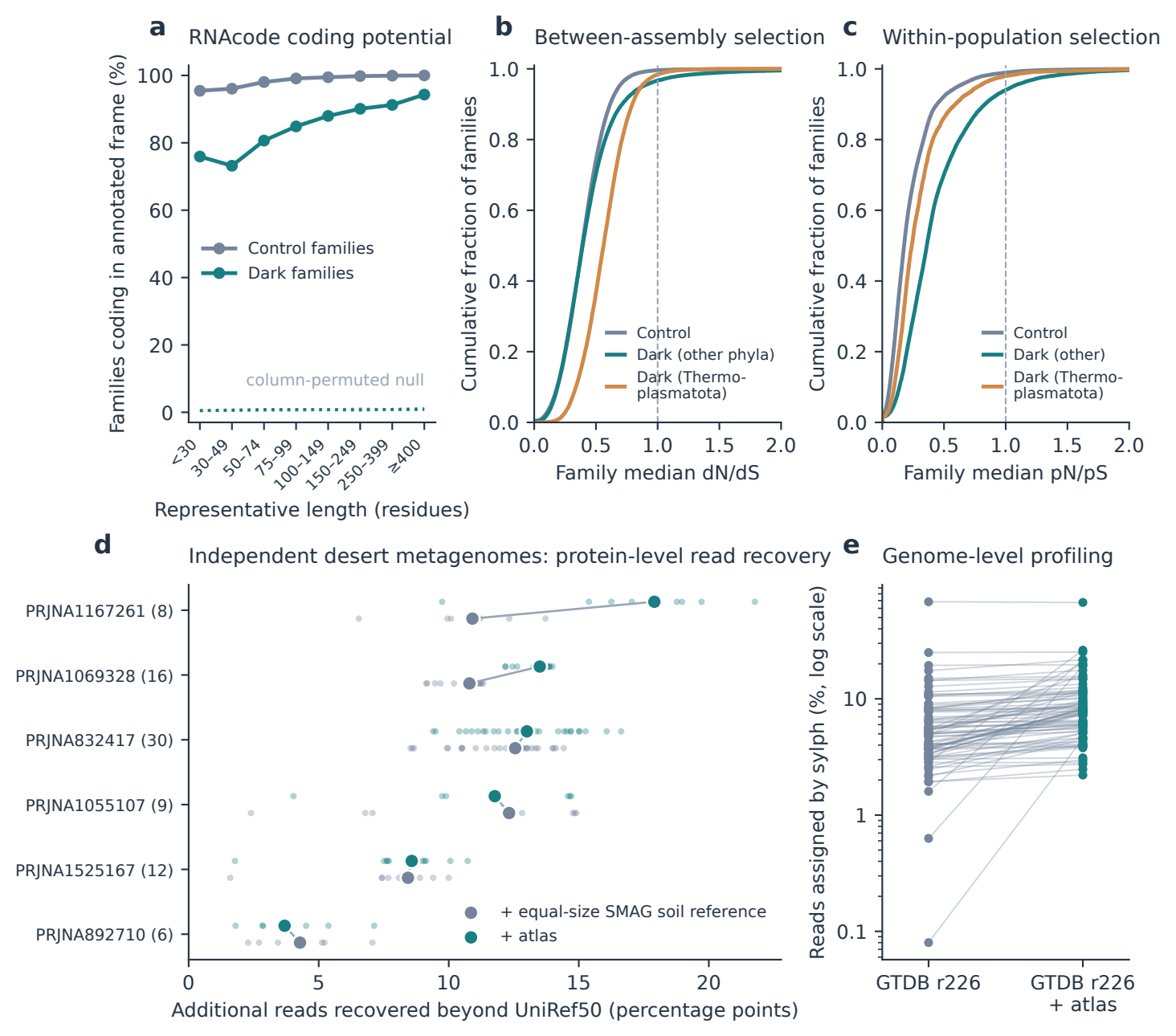

a
RNAcode coding potential
Families coding in annotated frame (%)
100
80
60
40
20
0
Control families
Dark families
column-permuted null
<30
30–49
50–74
75–99
100–149
150–249
250–399
≥400
Representative length (residues)
b
Between-assembly selection
Cumulative fraction of families
1.0
0.8
0.6
0.4
0.2
0.0
Control
Dark (other phyla)
Dark (Thermo-plasmatota)
0.0
0.5
1.0
1.5
2.0
Family median dN/dS
c
Within-population selection
Cumulative fraction of families
Control
Dark (other)
Dark (Thermo-plasmatota)
Family median pN/pS
d
Independent desert metagenomes: protein-level read recovery
PRJNA1167261 (8)
PRJNA1069328 (16)
PRJNA832417 (30)
PRJNA1055107 (9)
PRJNA1525167 (12)
PRJNA892710 (6)
+ equal-size SMAG soil reference
+ atlas
0
5
10
15
20
Additional reads recovered beyond UniRef50 (percentage points)
e
Genome-level profiling
Reads assigned by sylph (%, log scale)
10
1
0.1
GTDB r226
GTDB r226 + atlas